

What is Deductive reasoning?

"Deductive reasoning is a basic form of valid reasoning. Deductive reasoning, or deduction, starts with a general statement or hypothesis, and examines the possibilities to reach a specific, logical conclusion, according to California State University." (Bradford)

As one might imagine from the above information, deductive reasoning is used heavily in science. However, beyond that, one is likely to use deductive reasoning within one's daily life both at work on a fairly regular basis. For example, a manager of a company might notice a budget has been exceeded and deduces that travel expenses should be cut back to manage a business's finances better.

What is Inductive reasoning?

"Inductive reasoning is the opposite of deductive reasoning. Inductive reasoning makes broad generalizations from specific observations. There is data; then, conclusions are drawn from the data. This is called inductive logic, according to Utah State University." (Bradford)

Inductive reasoning can be compared to assumptions based on data. For example, pattern recognition. Consider a basket full of mangos, and you want to find out if they are raw or ripe, we pick up a mango and observe it, let say we find the mango to be raw, and we did the same with four mangos, and all of them are raw. We might use inductive reasoning to determine that all the mangos in the basket are raw.

Compare and contrast

However, both types of reasoning have a place in science, and their use cases are different. For example, you wouldn't use inductive reasoning if attempting to reach a specific conclusion, like $Y = 10$; this example would apply to deductive reasoning.

Inductive reasoning could be used to reach a generalized conclusion about an abstract concept, such as pattern recognition. Consider seals, and they swim north for the winter. How did we reach that conclusion? We observed the behaviour and determined that they migrate to a cold area, based on presently available data. Is the conclusion specific? No, it's a generalized statement. However, where they migrate uses deductive reasoning, because we are attempting to determine the why, where and when. Every type of reasoning is used in biological research.

2) I believe, as it relates to establishing a cause and the effect relationship, could be used in both deductive reasoning and inductive reasoning. At its core, a cause and effect relationship is an event that is related to or causes another event. As it relates to software, a cause and effect relationship might be a user deducing that clicking the red "X" circle at the top left hand of the screen (macOS), it will close the application or web page. In this example, the user hypothesizes that the "X" represents the action of

"Close" or "Exit." The user then tests this hypothesis by clicking on it. The web page closes. The cause and effect relationship has been established. This shows a straightforward example of deductive reasoning and cause and effect relationship.

Resources:

Bradford, A. (2017, July 25). Deductive Reasoning vs. Inductive Reasoning. Retrieved June 23, 2020, from <https://www.livescience.com/21569-deduction-vs-induction.html>